

Safety Walk-through

Covering SDS/MSDS

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If I have a question, where do I go?

What chemical is this?

What is in the product?

There is a fire, what matters?

How should we handle/store it?

How will it physically behave?

What are the health effects?

How should it be disposed?

What regulations apply?

What are the hazards?

Someone was exposed, what now?

There is a spill, what do we do?

What exposure controls/ PPE?

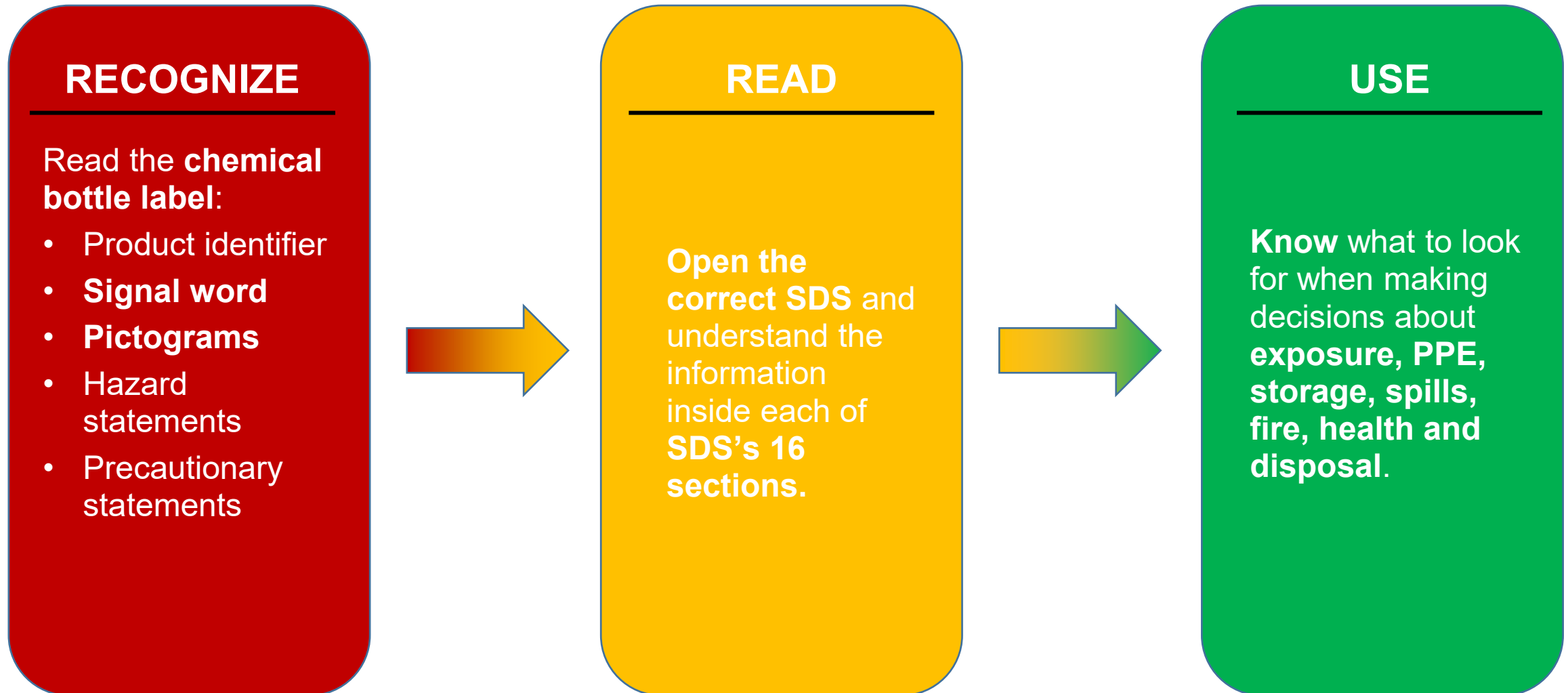
What can it react with?

Environmental impact?

How is it transported?

Is this document current?

What this training is designed to teach



Presented By abusufiyan

Step 1: Start with the chemical bottle

The label is the rapid hazard-recognition layer.

Product identifier: Product name, CAS No or product code used to **identify the chemical** and find the correct SDS.

Signal word: Shows **how serious** the hazard is:
DANGER or WARNING.

Pictograms: Symbols that quickly show the **type of hazard**.

Hazard statements: Explains **what harm** the chemical can cause.

Precautionary statements: Explains how to prevent exposure, what to do if exposed, and how to store or dispose of it safely.

Supplier information: Shows who supplied the chemical and who to contact for more information.

● What information to extract form the Label

match it to the SDS.

CODE _____
Product Name _____

Company Name _____
 Street Address _____
 City _____ State _____
 Postal Code _____
 Emergency Phone Number _____

SAMPLE LABEL

Hazard Pictograms



Signal Word

Danger

Hazard Statements

Highly flammable liquid and vapor.
May cause liver and kidney damage.

Product Identifier

Supplier Identification

Precautionary Statements

Supplemental Information

Directions for Use

Fill weight: _____ Lot Number: _____

Gross weight: _____ Fill Date: _____

Expiration Date: _____

Step 2: Find the correct SDS/ MSDS

The product identifier on the label should connect you to the correct SDS.

What is SDS/MSDS ?

- **SDS:** Safety Data Sheet or **MSDS:** Material Safety Data Sheet
- A standardized safety reference containing information about the chemical's identity, hazards, composition, emergency measures, controls, properties, health effects and other information.
- **BEFORE USING IT:**
 - Check:
 - ✓ Product identifier match it to label of chemical bottle
 - ✓ Supplier/manufacturer
 - ✓ Product grade/formulation when relevant
 - ✓ Revision date
 - ✓ Whether it is the current SDS

Think of MSDS/ SDS as the detailed safety reference behind the label.

Never go for the Generic SDS when Vendor SDS is available. Always priorities SDS by Manufacturer.

Step 2(a): The 16-section SDS map

● Main Focus

The sections follow a standardized sequence.

SAFETY DATA SHEET

Creation Date :XX-XX-XXXX

Revision Date :XX-XX-XXXX

Section 1: Identification

Section 2: **Hazard(s) Identification**

Section 3: Composition/Information on Ingredients

Section 4: **First-Aid Measures**

Section 5: **Fire-Fighting Measures**

Section 6: **Accidental Release Measures**

Section 7: **Handling and Storage**

Section 8: Exposure Controls/Personal Protection

Emergency
Response &
Control
Measures

Required by
GHS format .
Not enforced
by OSHA

Section 9: Physical and Chemical Properties

Section 10: Stability and Reactivity

Section 11: **Toxicological Information**

Section 12: Ecological Information

Section 13: Disposal Considerations

Section 14: Transport Information

Section 15: Regulatory Information

Section 16: Other Information

Section 1: Identification

SAFETY DATA SHEET

Creation Date :XX-XX-XXXX

Revision Date :XX-XX-XXXX

Section 1: Identification

Section 2: Hazard(s) Identification

Section 3: Composition/Information on Ingredients

Section 4: First-Aid Measures

Section 5: Fire-Fighting Measures

Section 6: Accidental Release Measures

Section 7: Handling and Storage

Section 8: Exposure Controls/Personal Protection

WHAT YOU WILL FIND

- Product identifier used on the label
- Other means of identification
- Recommended use and restrictions on use
- Manufacturer/importer/re

WHY IS IT THERE?

- It answers: “Exactly what product am I dealing with, what is it intended for, and who can provide authoritative help?”

WORKER FOCUS

- Match the bottle to the SDS before using it. Check supplier and emergency contact

Section 2: Hazard(s) Identification

SAFETY DATA SHEET

Creation Date :XX-XX-XXXX

Revision Date :XX-XX-XXXX

Section 1: Identification

Section 2: Hazard(s) Identification

Section 3: Composition/Information on Ingredients

Section 4: First-Aid Measures

Section 5: Fire-Fighting Measures

Section 6: Accidental Release Measures

Section 7: Handling and Storage

Section 8: Exposure Controls/Personal Protection

WHAT YOU WILL FIND

- Hazard classification
- Signal word
- Hazard statements
- Pictograms / symbols
- Precautionary statements
- Hazards not otherwise classified, where identified

WHY IS IT THERE?

- It is the SDS hazard summary.
- It answers: “What are the classified hazards and what are the main precautions?”

WORKER FOCUS

- Read the statements, not only the pictograms. Compare Section 2 with the container label.

Section 3: Composition / Information on Ingredients

SAFETY DATA SHEET

Creation Date :XX-XX-XXXX

Revision Date :XX-XX-XXXX

Section 1: Identification

Section 2: Hazard(s) Identification

Section 3: Composition/Information on Ingredients

Section 4: First-Aid Measures

Section 5: Fire-Fighting Measures

Section 6: Accidental Release Measures

Section 7: Handling and Storage

Section 8: Exposure Controls/Personal Protection

WHAT YOU WILL FIND

- Chemical name
- Common name / synonyms
- CAS number or other identifiers
- Relevant impurities/stabilizers
- For mixtures: hazardous ingredients and concentration or concentration ranges, subject to applicable rules/trade-secret provisions

WHY IS IT THERE?

- It answers: “What substances are responsible for the product's hazards?”

WORKER FOCUS

- Useful when evaluating a mixture, exposure, compatibility, toxicology and regulatory questions.

Section 4: First-Aid Measures

SAFETY DATA SHEET	
Creation Date :XX-XX-XXXX	Revision Date :XX-XX-XXXX
Section 1: Identification	
Section 2: Hazard(s) Identification	
Section 3: Composition/Information on Ingredients	
Section 4: First-Aid Measures	
Section 5: Fire-Fighting Measures	
Section 6: Accidental Release Measures	
Section 7: Handling and Storage	
Section 8: Exposure Controls/Personal Protection	

WHAT YOU WILL FIND

Necessary first-aid measures by route of exposure

- Inhalation
- Skin contact
- Eye contact
- Ingestion
- Most important symptoms/effects
- Indication of immediate medical attention and special treatment, when necessary

WHY IS IT THERE?

- It answers: “What should be done immediately after exposure?”

WORKER FOCUS

- Use the actual product SDS and workplace emergency/medical procedure. Do not improvise treatment from a training example.

Section 5: Fire-Fighting Measures

SAFETY DATA SHEET

Creation Date :XX-XX-XXXX

Revision Date :XX-XX-XXXX

Section 1: Identification

Section 2: Hazard(s) Identification

Section 3: Composition/Information on Ingredients

Section 4: First-Aid Measures

Section 5: Fire-Fighting Measures

Section 6: Accidental Release Measures

Section 7: Handling and Storage

Section 8: Exposure Controls/Personal Protection

WHAT YOU WILL FIND

- Suitable extinguishing media
- Unsuitable extinguishing media, where relevant
- Specific hazards arising from the chemical
- Hazardous combustion products
- Special protective equipment and precautions for firefighters

WHY IS IT THERE?

- It answers: “If there is a fire, what hazards and firefighting controls are relevant?”

WORKER FOCUS

- Look beyond “flammable.” Combustion products and unsuitable extinguishing media can also matter.

Section 6: Accidental Release Measures

SAFETY DATA SHEET

Creation Date :XX-XX-XXXX

Revision Date :XX-XX-XXXX

Section 1: Identification

Section 2: Hazard(s) Identification

Section 3: Composition/Information on Ingredients

Section 4: First-Aid Measures

Section 5: Fire-Fighting Measures

Section 6: Accidental Release Measures

Section 7: Handling and Storage

Section 8: Exposure Controls/Personal Protection

WHAT YOU WILL FIND

- Personal precautions
- Protective equipment
- Emergency procedures
- Methods/materials for containment
- Methods/materials for cleaning up

WHY IS IT THERE?

- It answers: “What should we do if the chemical spills or is accidentally released?”

WORKER FOCUS

- Separate the response problem from normal handling. Use site spill procedures and the actual SDS.

Section 7: Handling and Storage

SAFETY DATA SHEET

Creation Date :XX-XX-XXXX

Revision Date :XX-XX-XXXX

Section 1: Identification

Section 2: Hazard(s) Identification

Section 3: Composition/Information on Ingredients

Section 4: First-Aid Measures

Section 5: Fire-Fighting Measures

Section 6: Accidental Release Measures

Section 7: Handling and Storage

Section 8: Exposure Controls/Personal Protection

WHAT YOU WILL FIND

- Precautions for safe handling
- Conditions for safe storage
- Incompatibilities
- Other storage conditions where relevant

WHY IS IT THERE?

- It answers: “It converts hazard information into routine controls that help prevent an incident.”

WORKER FOCUS

- Think: “How do I prevent a fire, exposure, reaction or deterioration during normal work and storage?”

Section 8: Exposure Controls/Personal Protection

SAFETY DATA SHEET

Creation Date :XX-XX-XXXX

Revision Date :XX-XX-XXXX

Section 1: Identification

Section 2: Hazard(s) Identification

Section 3: Composition/Information on Ingredients

Section 4: First-Aid Measures

Section 5: Fire-Fighting Measures

Section 6: Accidental Release Measures

Section 7: Handling and Storage

Section 8: Exposure Controls/Personal Protection

WHAT YOU WILL FIND

- Occupational exposure limits where available
- Appropriate engineering controls
- Individual protection measures
- Eye/face protection
- Skin/hand protection
- Respiratory protection where applicable

WHY IS IT THERE?

- It answers: “How do we control exposure, and what protection is needed for the task?”

WORKER FOCUS

- Engineering controls come first in the hierarchy of controls; PPE is selected based on the actual hazard and task.

Section 9: Physical and Chemical Properties

WHAT YOU WILL FIND

- Appearance / physical state / color
- Odor and odor threshold, pH
- Melting/freezing point, Boiling point / range, Flash point, Evaporation rate
- Flammability / explosive limits, Vapor pressure / density
- Solubility, Partition coefficient, Viscosity
- Auto-ignition / decomposition temperature

WHY IS IT THERE?

- It answers: “These properties help predict how the chemical behaves during use, storage and emergencies.”

WORKER FOCUS

- Don't memorize numbers. Ask what each property means for volatility, fire, spill behavior, storage or handling.

Section 9: Physical and Chemical Properties

Section 10: Stability and Reactivity

Section 11: Toxicological Information

Section 12: Ecological Information

Section 13: Disposal Considerations

Section 14: Transport Information

Section 15: Regulatory Information

Section 16: Other Information

Section 10: Stability and Reactivity

WHAT YOU WILL FIND

- Reactivity
- Chemical stability
- Possibility of hazardous reactions
- Conditions to avoid
- Incompatible materials
- Hazardous decomposition products

WHY IS IT THERE?

- It answers: “What can make this chemical react unexpectedly or dangerously?”

WORKER FOCUS

- Critical for mixing, storage compatibility, heat, ignition sources, contamination and emergency response.

Section 9: Physical and Chemical Properties

Section 10: Stability and Reactivity

Section 11: Toxicological Information

Section 12: Ecological Information

Section 13: Disposal Considerations

Section 14: Transport Information

Section 15: Regulatory Information

Section 16: Other Information

Section 11: Toxicological Information

WHAT YOU WILL FIND

- Likely routes of exposure
- Symptoms related to physical/chemical/toxicological characteristics
- Immediate and delayed effects
- Chronic effects from short- and long-term exposure
- Numerical measures of toxicity
- Relevant carcinogenicity information where applicable

WHY IS IT THERE?

- It answers: “It provides the detailed health-effects picture behind the hazard classification.”

WORKER FOCUS

- Use it to understand the health significance of exposure routes and effects; do not treat one number as the whole risk assessment.

Section 9: Physical and Chemical Properties

Section 10: Stability and Reactivity

Section 11: Toxicological Information

Section 12: Ecological Information

Section 13: Disposal Considerations

Section 14: Transport Information

Section 15: Regulatory Information

Section 16: Other Information

Section 12: Ecological Information

WHAT YOU WILL FIND

- Eco-toxicity
- Persistence and degradability
- Bio-accumulative potential
- Mobility in soil
- Other adverse effects, where relevant

WHY IS IT THERE?

- It answers: “What can this chemical do to the environment if it is released?”

WORKER FOCUS

- Important for spill, waste and environmental decision-making. Requirements vary by jurisdiction.

Section 9: Physical and Chemical Properties

Section 10: Stability and Reactivity

Section 11: Toxicological Information

Section 12: Ecological Information

Section 13: Disposal Considerations

Section 14: Transport Information

Section 15: Regulatory Information

Section 16: Other Information

Section 13: Disposal Considerations

WHAT YOU WILL FIND

- Waste residues
- Safe handling of waste
- Methods of disposal
- Contaminated packaging

WHY IS IT THERE?

- It answers: “It helps identify disposal considerations so the chemical and contaminated packaging are managed appropriately.”

WORKER FOCUS

- The SDS does not override local waste laws, permits or your site's waste procedure.

Section 9: Physical and Chemical Properties

Section 10: Stability and Reactivity

Section 11: Toxicological Information

Section 12: Ecological Information

Section 13: Disposal Considerations

Section 14: Transport Information

Section 15: Regulatory Information

Section 16: Other Information

Section 14: Transport Information

WHAT YOU WILL FIND

- UN number
- UN proper shipping name
- Transport hazard class(es)
- Packing group, if applicable
- Environmental hazards
- Special transport precautions

WHY IS IT THERE?

- It answers: “What information is needed when the chemical is transported?”

WORKER FOCUS

Use the current product classification and the rules applicable to the mode and jurisdiction of transport.

Section 9: Physical and Chemical Properties

Section 10: Stability and Reactivity

Section 11: Toxicological Information

Section 12: Ecological Information

Section 13: Disposal Considerations

Section 14: Transport Information

Section 15: Regulatory Information

Section 16: Other Information

Section 15: Regulatory Information

WHAT YOU WILL FIND

- Safety, health and environmental regulations specific to the product
- Other regulatory information required by the applicable jurisdiction

WHY IS IT THERE?

- It answers: “It identifies regulatory information specific to the product and jurisdiction.”

WORKER FOCUS

- Do not assume a regulatory statement from one country applies in another.

Section 9: Physical and Chemical Properties

Section 10: Stability and Reactivity

Section 11: Toxicological Information

Section 12: Ecological Information

Section 13: Disposal Considerations

Section 14: Transport Information

Section 15: Regulatory Information

Section 16: Other Information

Section 16: Other Information

WHAT YOU WILL FIND

- Date of preparation
- Date of last revision
- Other useful information
- References / abbreviations / notes where provided

WHY IS IT THERE?

- It answers: “It gives context needed to judge how current the SDS is and understand additional notes or references.”

WORKER FOCUS

- Always check the revision date when deciding whether the document is current.

Section 9: Physical and Chemical Properties

Section 10: Stability and Reactivity

Section 11: Toxicological Information

Section 12: Ecological Information

Section 13: Disposal Considerations

Section 14: Transport Information

Section 15: Regulatory Information

Section 16: Other Information

If I have a question, where do I go?

What chemical is this? : Section 1: Identification

What is in the product? : Section 3: Composition

There is a fire, what matters? :Section 5: Fire-Fighting

How should we handle/store it? : Section 7:Handling & Storage

How will it physically behave? :Section 9: Properties

What are the health effects? : Section 11: Toxicology

How should it be disposed? : Section 13: Disposal

What regulations apply? : Section 15: Regulatory Information

What are the hazards? :Section 2: Hazard

Someone was exposed, what now? :Section 4: First-Aid Measures

There is a spill, what do we do? : Section 6: Accidental Release

What exposure controls/ PPE? : Section 8: Exposure Control

What can it react with? : Section 10: Stability and Reactivity

Environmental impact? : Section 12: Ecological Information

How is it transported? : Section 14: Transport Information

Is this document current? : Section 16: Other Information

GHS Hazard Pictogram, H Code, **P Code, NFPA and PPE**

Hazard Communication Standard Pictogram



GHS01:
Explosive



GHS02:
Flammable



GHS03:
Oxidizing



GHS04:
Compressed Gas



GHS05:
Corrosive



GHS06: Toxic



GHS07: Health
Hazard/Hazardous
to Ozone Layer



GHS08: Serious
Health Hazard



GHS09:
Hazardous to the
Environment

Hazard Communication Standard Pictogram

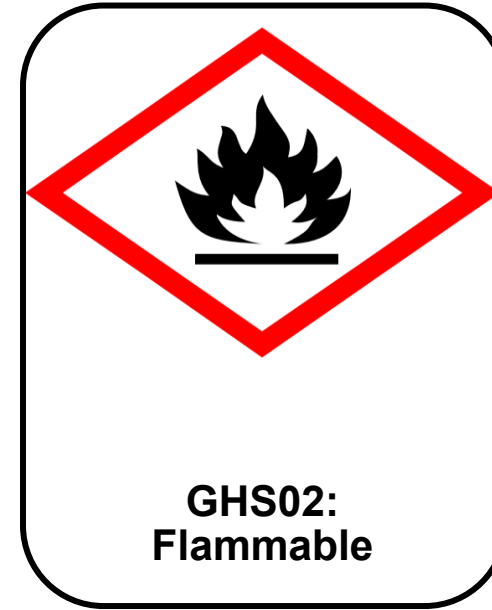


GHS01: Explosives

- Organic Peroxide
- Self-Reactives

Further information: [Explosive material](#)

e.g. azidoazide azide, TNT



GHS02: Flammables

- Pyrophorics, Organic Peroxides
- Self-Heating, Self-Reactives
- Emits Flammable Gas, Desensitized Explosive

Further information: [Flammability](#)

e.g. acetone, methanol, most of solvent

Hazard Communication Standard Pictogram

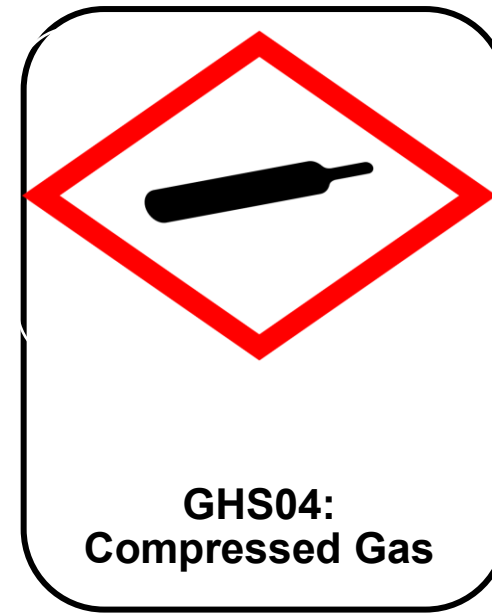


GHS03: Oxidizing

- Oxidizing gases,
- Oxidizing liquids
- Oxidizing solids

Further information: [Oxidizing agent](#)

e.g. hydrogen peroxide, most halogens, nitric acid

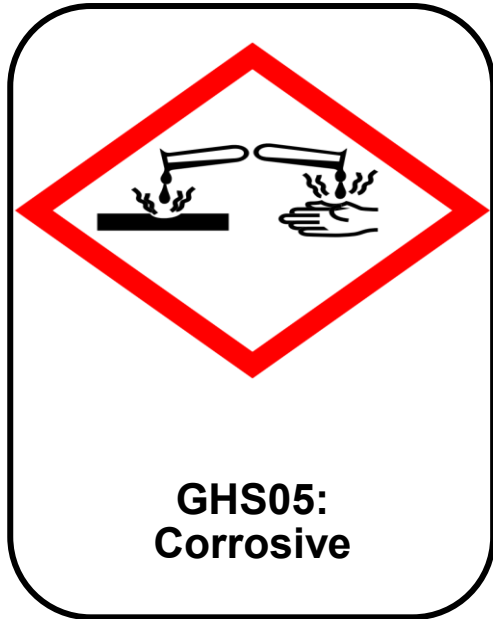


GHS04: Compressed Gas

- Compressed gases
- Liquefied gases, sRefrigerated liquefied gases
- Dissolved gases

e.g. liquid nitrogen, liquid oxygen, liquid helium

Hazard Communication Standard Pictogram



GHS05: Corrosive

- Skin Corrosion/Burns, Eye Damage
- Corrosive to Metals

Further information: [Corrosive substance](#)

e.g. Strong acids/bases (nitric acid), calcium oxide



GHS06: Toxic

- Acute Toxicity
 - (fatal or toxic)

e.g. manganese heptoxide

Hazard Communication Standard Pictogram



GHS07: Health Hazard/Hazardous to Ozone Layer

- Irritant (skin and eye)
- Skin Sensitizer
- Acute Toxicity (harmful)
- Narcotic Effects



GHS08: Serious Health Hazard

- Carcinogen
- Mutagenicity
- Reproductive Toxicity
- Target Organ Toxicity, Aspiration Toxicity

e.g. chromium, cadmium

Hazard Communication Standard Pictogram



GHS09: Hazardous to the Environment

- Aquatic Toxicity
- Chronic hazards to the aquatic environment

For detailed info **GHS Hazard Statements**,
H Code and **P Code**, please refer to :
<https://pubchem.ncbi.nlm.nih.gov/ghs/>

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GHS Classification (Rev.11, 2025) Summary

GHS, the Globally Harmonized System of Classification and Labeling of Chemicals, was developed by the United Nations as a way to bring into agreement the chemical regulations and standards of different countries. GHS includes criteria for the classification of health, physical and environmental hazards, as well as specifying what information should be included on labels of hazardous chemicals as well as safety data sheets. This page summarizes the relationship of GHS hazard statements, pictograms, signal words, hazard classes, categories, and precautionary statements.

References:

1. UNECE GHS (Rev.11, 2020)
2. UNECE GHS (Rev.10, 2023)
3. UNECE GHS (Rev.9, 2021)
4. UNECE GHS (Rev.8, 2019)
5. UNECE GHS (Rev.7, 2017)
6. UNECE GHS (Rev.6, 2015)
7. UNECE GHS (Rev.5, 2013)

Note: This page provides the current GHS summary. Obsolete (marked as deleted in GHS Rev.11 2025 PDF document) H-codes and P-codes are also provided, as they are still in use but annotated as obsolete. For more information, please see the References. This is the permanent link to the GHS 2025 (Rev. 11) summary. Please also see the previous GHS 2023 (Rev. 10) summary, GHS 2021 (Rev. 9) summary and GHS 2019 (Rev. 8) summary.

Download: The GHS summary data file (TSV).

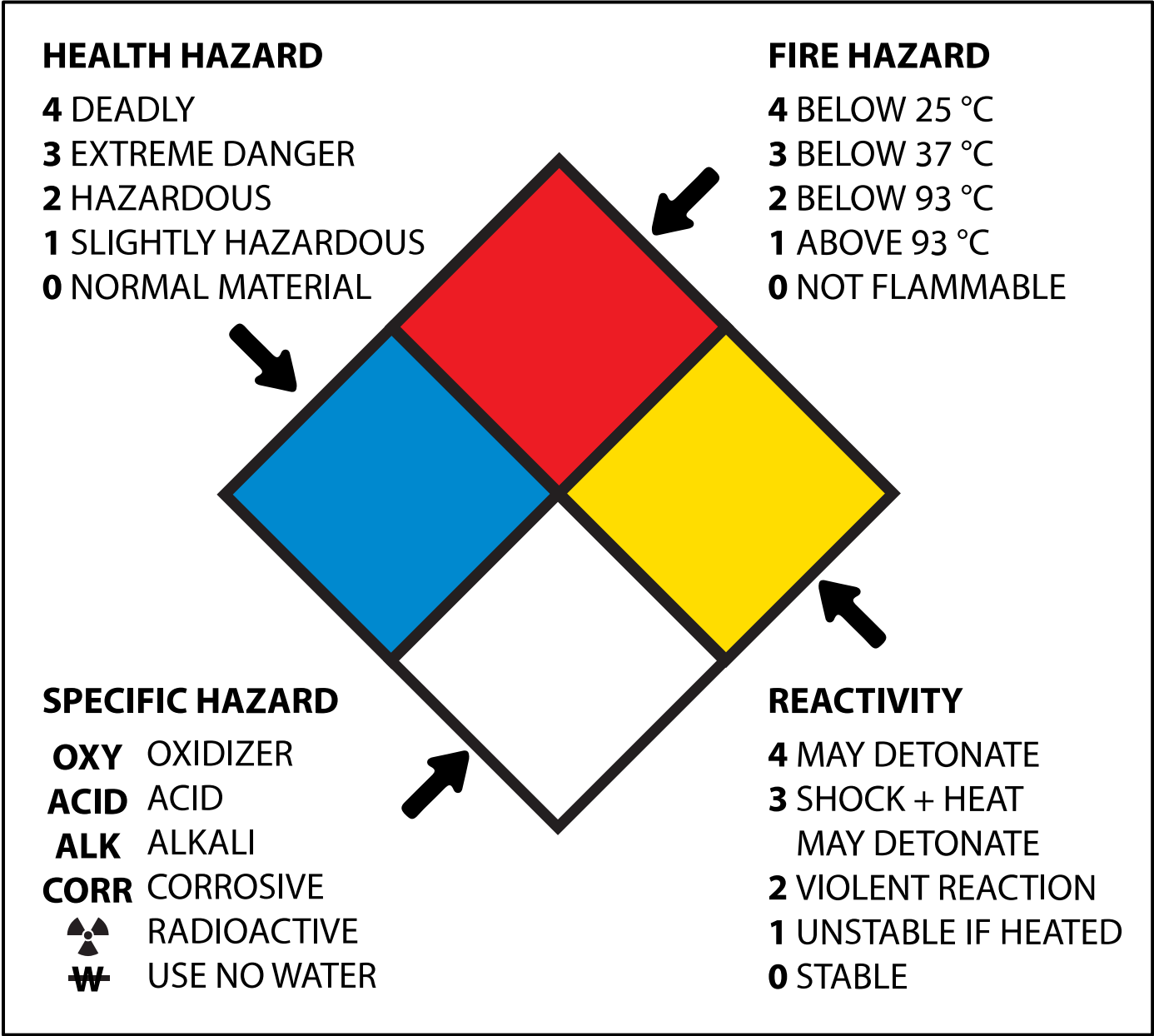
Hazard Class Pictograms

Exploding Bomb Explosives GHS01	Flame Flammables GHS02	Gas Cylinder Compressed Gases GHS04
Corrosion Corrosives GHS05	Skull and Crossbones Acute Toxicity GHS06	Exclamation Mark Irritant GHS07
Health Hazard GHS08	Environment GHS09	

GHS Hazard Statements

H-Code	Hazard Statements	Hazard Class	Category	Pictogram	Signal Word	UN Model Regulations Class or Division	UN Model Placed	Precautionary Prevention	Precautionary Response	Precautionary Storage	Precautionary Disposal
H200	(Extremely) Unstable Explosive	Explosives	Unstable Explosive		Danger			P201, P202, P281	P372, P373, P380	P401	P501
H201	(Extremely) Explosive, mass explosive hazard	Explosives			Danger	Div 1.1		P201, P202, P281, P283, P284	P372, P373, P380	P401	P501

NFPA diamond



PPE (Personal Protective Equipment)

Don't memorize 16 definitions, learn the questions the SDS answers.

Eye & face protection : Protects against splashes, sparks, and flying particles.

Skin protection : Protects against harmful chemical contact.

Hand protection : Use suitable chemical-resistant gloves.

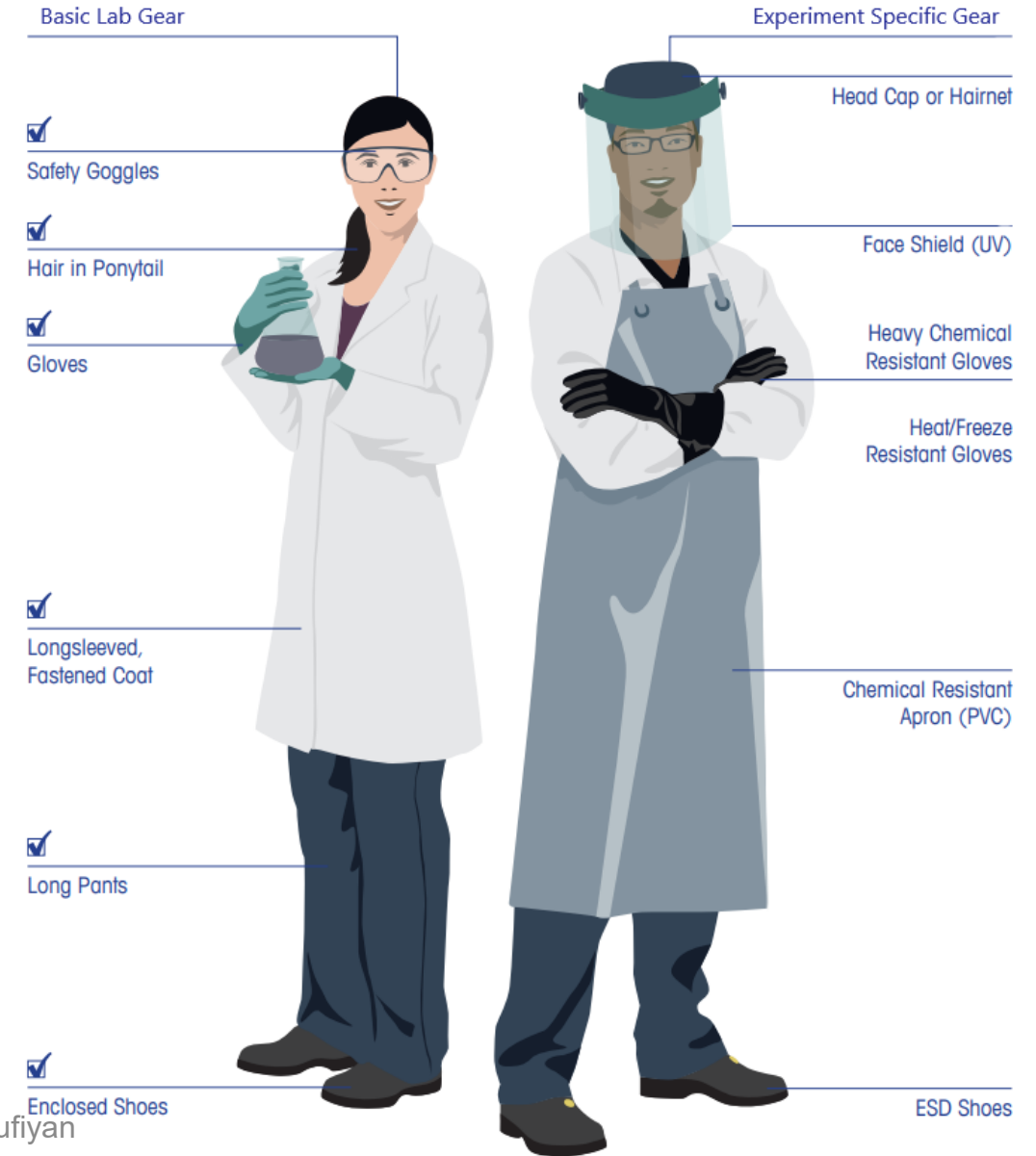
Respiratory protection : Protects against contaminated air.

Foot protection : Use chemical-resistant boots when required.

Hearing protection : Use earplugs in high-noise areas.

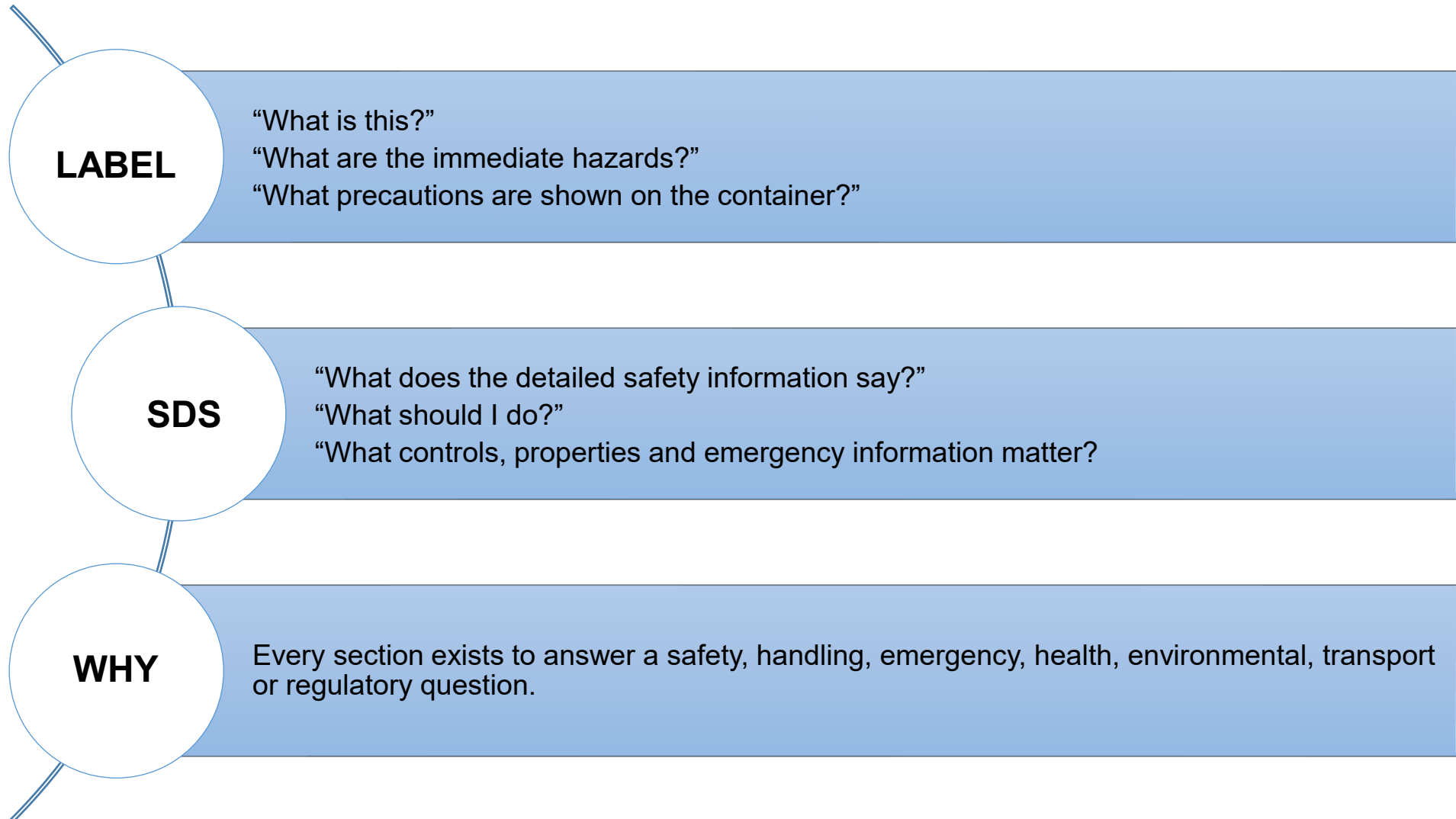
Select the right PPE : Choose PPE based on the hazard and exposure risk.

PPE has limits : It must be used correctly with other safety controls.



The mental model to remember

Don't memorize 16 definitions, learn the questions the SDS answers.



THANK YOU

Any Question ?